

Problem scoping:

Industrial Machine Failure Prediction

Step1: Use Case: Industrial Machine Failure Prediction

Factory management

- Notices that its machine stop working suddenly without warning.
- They want a system that tells them machine's risk level.

They approach an AI Team

- Ask to build an AI model/system to identify machines which are likely to fail without warning.
- Sounds very straight forward.

The solution statement is not a well-defined AI problem.

Conversation Between Factory Management & AI Team

- Try to understand problem

Initial statement-

- **Machines are failing without warning.**
- **Needs to uncover the actual problem.**
- **AI team ask themselves several questions**

Questions for stakeholders:

- **Factory management**

What problems did industry face?

What output would you like from the AI system?

- **Maintenance Department:**

How often do these failures occur?

What interventions currently exists.

What much time is available before intervention

What impacts does failure have financially and socially.

How many machines can the maintenance team service each week

- **Engineering Department:**

Do you collect any machine data?

- **Suppose company gets following results**
- Machines stop unexpectedly, causing production downtime and financial losses.
- Classifies machines on the basis of risks level:

Failure Rate Class	
0-25%	Healthy
26- 50%	Low Risk
51-75%	Medium Risk
76- 100%	High Risk

- Approximately 8–10 breakdowns occur each month.
- At least 24 hours before failure.
- Each breakdown causes 2–5 hours of downtime and costs approximately ₹50,000–₹1,00,000.
- Maintenance team can maintain 10 machines per month.
- company continuously collect data such as:
 - Temperature
 - Vibration
 - RPM
 - Torque
 - Power Consumption

Step2: Defining Problem Precisely.

- **Poor Problem Statement**

Predict Machines risk's level

- **Better approach**

Predicts the failure risk level of industrial machines at least 24 hours in advance using sensor data, thereby reducing downtime and financial losses.

- **Maintainers can intervene**

Who?

Factory managers, maintenance engineers, machine operators.

What?

Unexpected machine failures causing losses.

Where?

Manufacturing plants and industrial environments.

Why?

To reduce downtime, maintenance costs, and improve safety.

Step 3: To Justify Is AI really needed? Or can be solved without AI.

➤ The maintenance team first tries simple rules:

- If Temperature > 80°C → High Risk
- If Vibration > 2.5 → High Risk

But it may be possible that:

- Some machines fail even below 80°C.
- Some machines work fine above 80°C.
- Failure depends on many factors together, not just one sensor.
- Also, checking millions of sensor readings manually is not possible.

AI should only use when:

- Patterns are complex,
- Multiple factors interact.
- Manual rules are insufficient.

In our case:

Risk's levels are based on many factors:

- Temperature
 - Vibration
 - RPM (Rotation Per Minute)
 - Torque
 - Power Consumption
- Patterns are going to be complex, that's why AI is needed.

Step4: Identifying Stakeholders:

Stakeholders- Everyone who is affected by the Project.

1. **Maintenance Department:** Receives early warning and improve productivity.
2. **Factory Manager:** Reduces downtime and increases productivity.
3. **Machine Operators:** Get alerts about risky machines.
4. **Production Department:** Ensures smooth production and fewer interruptions.
5. **IT Department:** Maintain software, servers, and dashboards.

Step5: Machine Learning Foundations

- **Classification Problem:** Output is either it is Healthy, Low, Medium or High Risks.
- Solid example of **Multiclass Classification**.

- **Risk Scoring:**

Failure Rate	Class
0–25%	Healthy
26–50%	Low Risk
51–75%	Medium Risk
76–100%	High Risk
- **Ranking:**

Which 10 machines should maintain first.
This aligns better with business objective.

Step6: Success Matrix:

Defining the success matrix:

- This is critical.
- **Technical Matrix** which we consider:
 - Accuracy
 - Precision
 - Recall
 - F1- Score
 - Confusion Matrix

Step7: Defining Constraints:

Every AI Project works under constraints

Constraint Types:

1. Operational constraints:

- Maintainers can maintain only 10 machines per month.

2. Time constraints:

- The system must provide warning at least 24 hours before failure.

3. Data Constraints:

- The AI model can only use the data collected by the company.
- If sensor data is missing or incomplete, prediction accuracy may decrease.

4. Ethical Constraints:

- Is the model should not be biased against a particular group.
- Unnecessary machines should not be maintained.
- Can predictions stigmatize machines.
- AI Prediction can never compromise employee safety.

Step8: Assumptions and Risks:

Assumptions:

- Historical data is available.
- Sensor's data are accurate.
- Maintenance Team will act on AI alerts.

Risks:

- Missing Data
- Historical Decisions
- Delayed Maintenance Actions.
- Unexpected Events that affect failure patterns.

Target Variables[Y]-

Machine Risk Level

Failure Rate (%) Target Class Numerical value

0–25%	Healthy	0
26–50%	Low Risk	1
51–75%	Medium Risk	2
76–100%	High Risk	3

Features Selection[X]-

1. Sensor Features-
 - Temperature
 - Vibration
 - RPM
 - Torque
 - Power Consumption
2. Machine Features-
 - Machine Age
 - Operation Hours
 - Tool Wear
 - Number of Previous Failures
3. Maintenance Features
 - Ambient Temperature
 - Humidity
4. Environmental Features
 - Ambient Temperature
 - Humidity

